

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
“Jnana Sangama”, Belgaum-590018.



PROJECT WORK PHASE-I REPORT

ON

**CrisisCore - A Real-Time AI Platform for Disaster
Preparedness, Response & Coordination**

*Submitted in Partial fulfilment for the VI Semester, BE, Artificial Intelligence & Data
Science*

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CERTIFICATE

Certified that the project work phase-I [23AI7PWPP1] entitled “**CrisisCore - A Real-Time AI Platform for Disaster Preparedness, Response & Coordination**” carried out by **Aastha Karn, Mounika B M, Sherisha C V** and **Vangireddy Anuthya**, bonafide students of BMS College of Engineering, in partial fulfillment for the award of Bachelor of Engineering in Artificial Intelligence & Data Science of the Visvesvaraya Technological University, Belgaum during the year 2025-26. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library.

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ABSTRACT

Natural disasters cause significant loss of life and infrastructure, particularly in developing regions. Existing disaster management systems are often fragmented, and lack real-time coordination among government agencies, NGOs, and rescue teams. The absence of a unified platform for situational awareness, predictive analytics, and coordinated response severely limits the ability to reduce casualty and economic damage during disaster events.

This project proposes CrisisCore – a Real-Time AI Platform for Disaster Preparedness, Response & Coordination. The system integrates machine learning models for disaster risk prediction using historical and real-time data (weather feeds, satellite imagery, and crowd-sourced reports). It incorporates a multilingual chatbot for citizen emergency reporting and guidance, priority detection algorithms for rescue operations, geospatial location tracking, and a centralized coordination dashboard for emergency responders.

The platform is designed with a scalable, open-platform architecture enabling seamless communication between government agencies, NGOs, hospitals, and citizens. Low-bandwidth and partial offline functionality ensure usability in rural and network-constrained environments. By enabling faster decision-making, optimized resource allocation, and data-driven situational awareness, CrisisCore aims to reduce disaster response time and strengthen overall disaster resilience.

DECLARATION

We, the undersigned students of the 6th Semester, Department of Artificial Intelligence & Data Science, BMS College of Engineering, declare that the Project Work Phase-I entitled

"CrisisCore – A Real-Time AI Platform for Disaster Preparedness, Response & Coordination"

is a bonafide work carried out by us. This project is neither a copy nor a modification of any other engineering project. We also declare that this project was not submitted to any other university in the past and will not be submitted to any other university in the future.

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Chapter 1

Introduction

1.1 Overview

Disasters, natural or man-made, represent one of the most persistent and complex challenges to sustainable development globally. According to the United Nations Office for Disaster Risk Reduction (UNDRR), between 2000 and 2019 alone, over 7000 disaster events were recorded worldwide, causing 1.23 million deaths, affecting 4.2 billion people, and resulting in economic losses of approximately USD 2.97 trillion.

India ranks among the top five most disaster-prone countries in the world. Its diverse geography exposes different regions to different hazards: the Gangetic plain and northeastern states face recurrent flooding; the eastern coastline is regularly struck by powerful cyclones; the Himalayan belt and parts of the Deccan plateau are vulnerable to earthquakes and landslides; and large swaths of central and western India experience prolonged droughts. The 2018 Kerala floods displaced over one million people and caused economic losses estimated at INR 40,000 crore. The 2023 Joshimath land subsidence crisis highlighted slow-onset urban disaster risks that existing monitoring systems were ill-equipped to detect and respond to proactively.

Despite advances in remote sensing, telecommunications, and computing, disaster response in India continues to suffer from three fundamental, systemic problems. First, information fragmentation: weather data, satellite imagery, field reports, and resource inventories exist in separate, incompatible databases maintained by different agencies (IMD, NDMA, state SDMAs, district collectors, NGOs, hospitals) with no unified operational view. Second, coordination failure: the process of mobilizing resources and communicating between agencies during an active disaster relies heavily on phone calls and ad-hoc messaging applications, prone to miscommunication, duplication of effort, and critical delays. Third, exclusion: citizens, particularly in rural and multilingual communities, have limited channels to report emergencies, receive guidance, or access early warning information. The absence of multilingual, low-bandwidth interfaces systematically disadvantages the populations most vulnerable to disaster.

The rapid advancement and increased accessibility of Artificial Intelligence, Machine Learning, Natural Language Processing, and geospatial analytics presents a transformative opportunity to address each of these systemic problems. AI models can cross-correlate large

volumes of diverse, multi-format data in real time, generating actionable situational awareness that would be impossible for human operators to construct manually under crisis conditions. NLP systems can process citizen-submitted text in regional languages, route emergency reports to appropriate authorities, and generate automated guidance. Computer vision models can assess structural damage from aerial imagery within minutes of satellite acquisition, eliminating bottlenecks in manual damage assessment. Integrated into a unified, purpose-built platform, these capabilities can dramatically reduce disaster response times, improve resource utilization, and save lives.

CrisisCore is conceived as precisely this unified platform. It is designed not as a research prototype for demonstration purposes, but as a practical, deployable system aligned with India's disaster management architecture, accessible to all stakeholders from national government agencies to individual rural citizens, and engineered to operate reliably under the very conditions disasters impose: poor network connectivity, high concurrent load, rapidly evolving situational information, and stressed human operators.

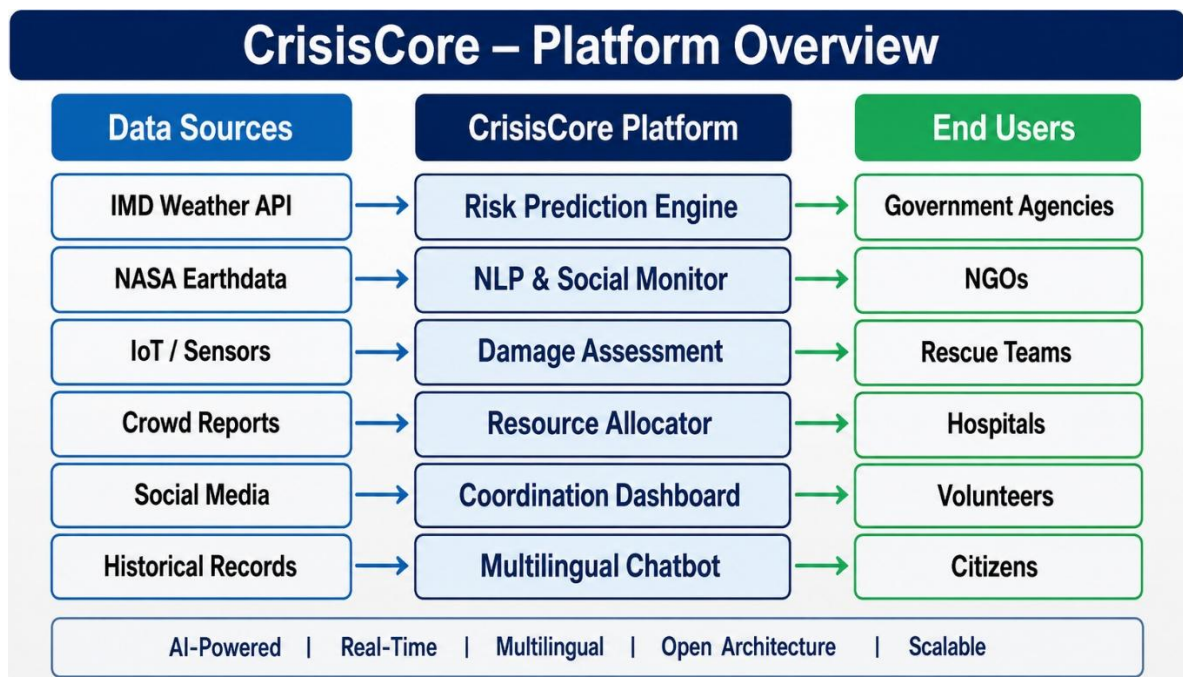


Figure 1.1: CrisisCore Platform Overview – Data Sources, Core AI Platform, and End Users

1.2 Motivation

The motivation for developing CrisisCore stems from several critical shortcomings observed in existing disaster management approaches:

- Lack of a unified platform: Government agencies, NGOs, rescue teams, and hospitals operate in silos, with no centralized system for real-time information sharing and coordination.
- Reactive rather than proactive systems: Most existing systems respond after a disaster has struck, rather than predicting risk zones and enabling preemptive action.
- Communication barriers in crisis zones: Rural and low-connectivity areas are severely underserved during disasters due to reliance on internet-dependent, English-only platforms.
- Manual and delayed decision-making: The absence of automated prioritization mechanisms leads to inefficient allocation of medical, food, and rescue resources.
- Inadequate damage assessment: Post-disaster damage quantification remains slow and manual, delaying insurance processing and reconstruction planning.

These gaps have been consistently highlighted in academic literature and disaster post-mortems [1][2][3]. CrisisCore is motivated by the vision of building a platform that addresses each of these challenges using state-of-the-art AI techniques.

1.3 Objectives

The primary objectives of the CrisisCore platform are:

1. To predict disaster risk zones and generate early-warning alerts using a machine learning ensemble trained on historical disaster records, real-time weather feeds, and environmental parameters including soil moisture and infrastructure vulnerability indices.
2. To provide continuous real-time situational awareness through integration of live data from meteorological APIs, satellite platforms, IoT sensor networks, and geotagged crowd-sourced field reports submitted through the platform and social media channels.
3. To perform automated, quantitative damage assessment within hours of a disaster event using convolutional neural network models applied to aerial and satellite imagery, producing district-level damage severity maps and affected area estimates.
4. To optimise resource allocation and rescue prioritisation through a transparent, auditable weighted scoring algorithm that accounts for incident severity, population density, resource availability, and geographic proximity.

5. To enable seamless multi-agency coordination through a centralised, role-differentiated real-time dashboard accessible across desktop and mobile devices, with live map visualisation and inter-agency communication tools.
6. To ensure inclusive access for all citizens — regardless of language, literacy, or connectivity — through a multilingual NLP chatbot that operates in Indian regional languages and maintains core functionality under low-bandwidth and partial-offline conditions.
7. To develop a modular, standards-compliant, open-architecture system deployable within the existing Indian disaster management framework, extensible to different disaster types, geographic regions, and organisational scales.

1.4 Scope

The scope of CrisisCore – Phase I encompasses the following:

- Design and architecture of the complete platform, including front-end, back-end, AI/ML module stack, and database schema.
- Literature survey covering AI-based disaster management, social media analytics, satellite-based damage assessment, and LLM-assisted crisis response.
- Requirements specification including functional requirements, non-functional requirements, hardware/software stack, cost estimates, and compliance standards.
- Prototype implementation of one core module (risk prediction).
- Alignment with national and international disaster management policies and AI ethics standards.

Phase II will cover full integration, testing across disaster scenarios, real-world API connections, and field evaluation.

1.5 Sustainable Development Goals (SDGs)

CrisisCore directly contributes to the following United Nations Sustainable Development Goals:

- SDG 11 – Sustainable Cities and Communities: Improving disaster resilience in urban and peri-urban areas through predictive early warning, coordinated response, and data-driven recovery planning reduces disaster mortality and displacement.

- SDG 13 – Climate Action: Integrating real-time climate and meteorological data into the prediction and response pipeline enables proactive responses to increasingly frequent climate-induced disaster events.
- SDG 3 – Good Health and Well-Being: Faster response times, better resource allocation (including medical supplies and hospital routing), and geospatially optimised rescue prioritisation directly reduce disaster-related morbidity and mortality.
- SDG 10 – Reduced Inequalities: Multilingual, low-bandwidth design ensures that rural, low-literacy, and non-English-speaking populations — who are typically most exposed to disasters and least served by existing systems — receive equal access to emergency services and guidance.
- SDG 17 – Partnerships for the Goals: A shared, open-architecture coordination platform facilitates effective inter-agency cooperation between national government, state authorities, NGOs, the private sector, and civil society.

1.6 Existing System and Limitations

Several platforms exist in the disaster management domain. A critical assessment reveals the specific gaps that CrisisCore is designed to fill.

1.6.1 NDMA Portal (India)

The National Disaster Management Authority web portal provides policy guidelines, district-level hazard profiles, historical disaster reports, and agency contact directories. While valuable as a reference and policy resource, it is entirely static — there are no operational tools for real-time coordination, live resource tracking, risk prediction, or citizen engagement during active disasters. It represents documentation of the disaster management system, not a tool for operating it.

1.6.2 Sahana Eden

Sahana Eden is a widely-deployed open-source Emergency Management platform used in responses to the Haiti earthquake (2010), Pakistan floods (2010), and other events. It provides modules for missing persons tracking, shelter management, and resource request handling. However, Sahana was designed for the post-disaster relief camp management phase — not real-time prediction or active response coordination. It has no AI/ML integration, no live data feeds, no NLP-based communication capabilities, and no predictive risk assessment. Its complexity makes rapid deployment challenging during the acute phase of a disaster.

1.6.3 Google Crisis Response & Person Finder

Google's crisis response tools provide real-time maps, public alert integrations, and a person finder system for locating missing individuals. These are citizen-facing information tools — they do not offer operational coordination capabilities, predictive analytics, automated damage assessment, or any mechanism for inter-agency resource management. They are supplements to official emergency management, not replacements.

1.6.4 FEMA Systems (USA)

The US Federal Emergency Management Agency has developed sophisticated internal coordination systems (IPAWS for wireless emergency alerts, WebEOC for situation reporting) over decades with substantial federal investment. These systems are deeply integrated with US law, infrastructure, and emergency management culture, making them inapplicable to the Indian context. They are also not open-source, and their licensing and infrastructure costs place them out of reach for most disaster-prone lower-income nations.

1.6.5 iSAFE

India's iSAFE satellite-based early warning platform provides hazard-specific alerts for floods, cyclones, and earthquakes using remote sensing data. While technically capable for its specific purpose, it is a single-function early warning tool with no citizen engagement, no resource management, no coordination dashboard, and no multi-agency integration capability. Its use requires specialised GIS training, limiting its operational utility.

In summary: no existing system — domestic or international — integrates disaster prediction, real-time multi-source data fusion, automated damage assessment, multilingual citizen engagement, and multi-agency operational coordination in a single, open, adaptable platform suited to the Indian disaster management context.

1.7 Proposed System

CrisisCore proposes a unified, AI-driven disaster management platform that addresses all the above gaps. The key distinguishing features of the proposed system are:

- Integrated AI/ML pipeline: Disaster risk prediction using Decision Trees, Random Forests, and Logistic Regression; NLP-based social media analytics using CrisisNLP datasets and Named Entity Recognition (NER) for location extraction.
- Centralized Real-Time Dashboard: A multi-role web dashboard for government agencies, NGOs, responders, and citizens with live data visualization, resource tracking, and alert management.

- **Multilingual Chatbot:** Supports emergency reporting and citizen assistance in multiple Indian languages with partial offline capability.
- **Damage Assessment Module:** Automated image analysis using OpenCV and TensorFlow for rapid post-disaster damage quantification.
- **Scalable Open Architecture:** Built on React.js (frontend), Node.js + Express (backend), PostgreSQL + PostGIS (spatial database), with standard APIs for extensibility.
- **Compliance-first Design:** Built in alignment with the Disaster Management Act (2005), DPDPA (2023), ISO 27001, and OWASP security standards.

1.8 Problem Statement

Existing disaster management systems are fragmented, reactive, and lack real-time data integration and coordination among agencies, resulting in delayed response, inefficient resource allocation, and limited accessibility for affected populations, thereby highlighting the need for a unified, intelligent platform that enables timely, data-driven, and inclusive disaster response.

1.9 Work Plan

The project is planned across the two semesters of the academic year 2026. . The complete timeline, showing Phase I and Phase II activities, is depicted in the Gantt chart (Figure 1.2) below.

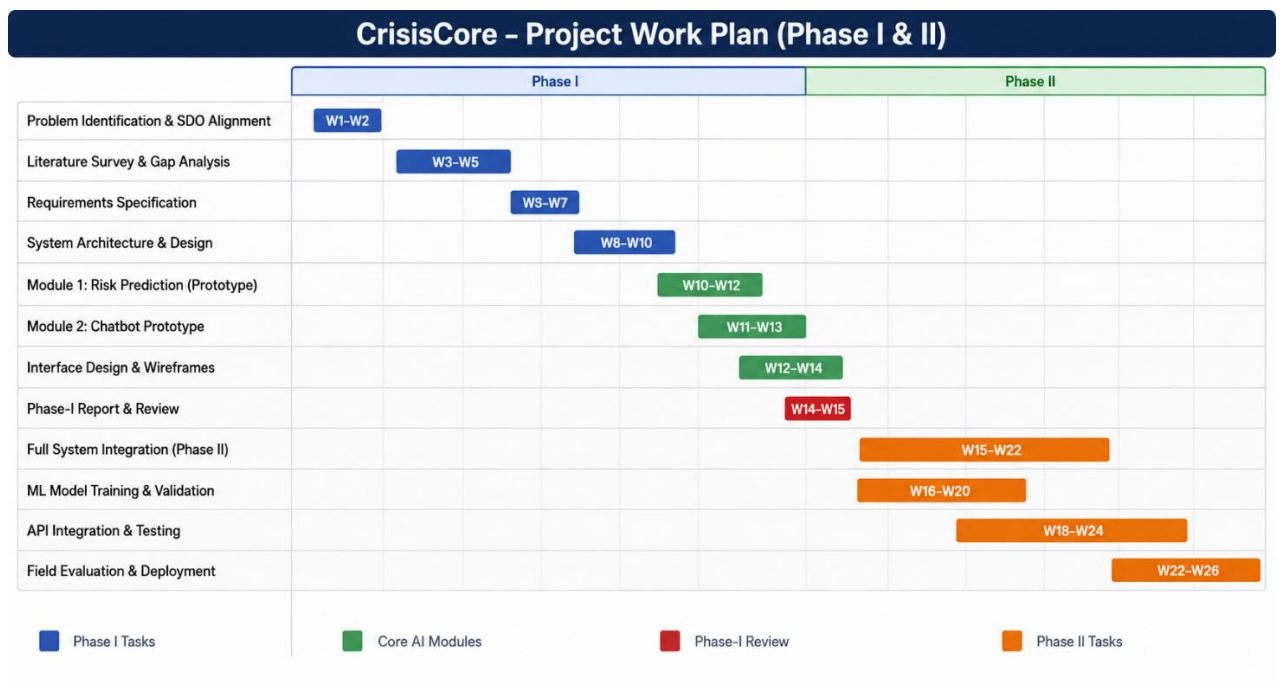


Figure 1.2: CrisisCore Project Work Plan – Gantt Chart (Phases I & II)

Phase I (Weeks 1–14): Problem identification and SDG alignment; exhaustive literature survey and gap analysis; complete requirements specification; system architecture and detailed design; prototype implementation of one core AI module; Phase-I report compilation and review. Phase II (Weeks 15–26): Full system integration across all five layers and eight modules; ML model training and rigorous evaluation on real disaster datasets; live API connections (IMD, NASA Earthdata, HDX); end-to-end testing using simulated disaster scenarios; user testing across all role categories; performance benchmarking; field evaluation with pilot agency; final report and deployment preparation.

Chapter 2

Literature Survey

2.1 Overview

A systematic literature survey was conducted to understand the state of the art in AI-driven disaster management, to identify the specific technical and functional gaps in existing research, and to establish the theoretical and empirical foundations for the CrisisCore platform. The survey covered peer-reviewed journal articles, conference proceedings, and technical reports published between 2020 and 2025, sourced from IEEE Xplore, ACM Digital Library, Nature, Google Scholar, MDPI, Springer, and the UNDRR knowledge repository. The review was structured around five thematic domains: (1) artificial intelligence and machine learning for disaster prediction and early warning; (2) social media analytics in crisis detection and situational awareness; (3) satellite-based and computer vision approaches to damage assessment; (4) large language model and NLP-based emergency management systems; and (5) multi-agency coordination platforms. Each domain is reviewed in narrative form below, identifying both the state of the art and the specific limitations that CrisisCore is designed to address.

2.2 Detailed Literature Survey

Table 2.1 below presents a structured comparison of the key papers surveyed, highlighting novelty, limitations, and gaps identified.

Table 2.1: Literature Survey – Detailed Comparison

Paper Title	Journal / Conference	Year	Novelty / Techniques	Limitations	Gap Identified
Social Media Analytics for Disaster Response: Classification and Geospatial Visualization Framework	Applied Sciences (MDPI)	2025	NLP tweet classification + geospatial visualization for real-time monitoring	Relies only on Twitter; location accuracy depends on user input; English only	No integration with emergency systems; lacks multilingual support and actionable coordination
LLM-Assisted Crisis Management for Emergency Response	IEEE Conference	2024	Uses LLAMA2 to classify emergency messages and generate response summaries	Works only in English; lacks integration with other data sources and live feeds	No routing, resource allocation, or real-time coordination capability

Mapping Global Floods with Satellite Radar Data	Nature Communications	2024	Deep learning with SAR imagery for automated flood extent detection	Limited to SAR data; resolution and timing constraints; no predictive capability	No integration with response systems or resource management modules
DisasterResponseGPT: LLMs for Disaster Response Planning	ICML Workshop	2023	Uses GPT-based LLMs to generate structured disaster response plans from textual inputs	Uses static inputs; no real-time data integration; no feedback loop	No dynamic adaptation, coordination, or real-time resource tracking
Flood Prediction Using Machine Learning: A Survey	IEEE Access	2023	Survey of CNN, LSTM, and ensemble methods for hydrological flood prediction	Survey paper – no novel system; models are region-specific and data-hungry	Limited generalization across different disaster types and geographies
AI-Based Early Warning Systems for Earthquake Preparedness	IJDRS	2022	Uses seismic sensor data with SVM and ANN for early warning generation	Requires expensive sensor infrastructure; high false-positive rates noted	No citizen communication layer; no integration with relief coordination systems

Beyond the tabulated papers, additional literature was reviewed in the areas of: volunteer coordination using mobile platforms [7], GIS-based resource optimization in emergency scenarios [8], federated learning for privacy-preserving disaster data sharing [9], and chatbot usability in emergency contexts [10]. These works collectively reinforce the need for a unified, multi-modal, real-time platform such as CrisisCore.

2.3 Research Gap Identification

Based on the literature survey, the following critical research gaps have been identified:

- No existing system integrates disaster prediction, real-time citizen reporting, damage assessment, and multi-agency coordination in a single unified platform.
- Multilingual support and low-bandwidth accessibility for rural populations remain largely unaddressed in current AI disaster management systems.

- Existing LLM-based approaches rely on static inputs and lack real-time data integration, limiting their operational utility during active disaster events.
- Social media and satellite-based systems operate independently without cross-domain data fusion for enhanced situational awareness.
- No open-source Indian-context platform addresses the full disaster lifecycle: preparedness, response, and recovery coordination.

2.4 Mapping of Objectives with Research Gaps

Table 2.2: Objective–Research Gap Mapping

CrisisCore Objective	Research Gap Addressed	References
ML-based disaster risk prediction	Lack of predictive capability in existing systems	[3][5][6]
Multilingual chatbot for citizen reporting	No multilingual or low-bandwidth access	[1][2][7]
Real-time multi-source data fusion	Siloed systems; no cross-domain integration	[1][3][4]
Centralized multi-agency coordination dashboard	Absence of unified coordination layer	[2][7][8]
AI-driven damage assessment via image analysis	No integration of satellite data with response	[3][5]

Chapter 3

Requirement Analysis and Specification

This chapter presents the complete requirements specification for the CrisisCore platform. Requirements were elicited through a structured process combining systematic literature analysis, national disaster management policy study, and reference to international humanitarian standards. Functional requirements define what the system must do; non-functional requirements define the quality constraints on how it must do it.

3.1 Functional Requirements

The functional requirements define the core capabilities that CrisisCore must provide. These are documented in Table 3.1.

Table 3.1: Functional Requirements

FR#	Feature	Description
FR1	Role-based User Management	Separate access roles for Admin, Government Agency, Rescue Team, NGO, Volunteer, and Citizen. Secure registration and authentication.
FR2	Disaster Risk Prediction	ML models predict high-risk zones using historical data, weather feeds, and environmental parameters. Outputs risk maps and early alerts.
FR3	Real-Time Data Integration	Ingests live data from IMD weather API, NASA satellite feeds, IoT sensors, and citizen-submitted reports.
FR4	Damage Assessment	Automated analysis of aerial/satellite imagery using OpenCV and CNN models to quantify damage extent.
FR5	Resource Allocation Dashboard	Centralized map-based dashboard showing resource availability (hospitals, shelters, rescue teams) with priority-based assignment.
FR6	Multilingual Chatbot	NLP-based chatbot for citizen reporting, guidance, and information in multiple Indian languages with partial offline mode.
FR7	Notification & Alert System	Automated alerts via in-app notifications and SMS for at-risk citizens and coordinating agencies.
FR8	Incident Reporting	Citizens and field teams can submit incident reports with location, severity, and media attachments.
FR9	Volunteer Coordination	Module for volunteer registration, skill mapping, and deployment to disaster zones.
FR10	Analytics & Reporting	Historical dashboards, post-disaster analytics, and exportable reports for agencies.

3.2 Non-Functional Requirements

Non-functional requirements define the system quality attributes that determine whether CrisisCore is fit for operational deployment under disaster conditions. These requirements are as critical as functional ones: a system that performs all the right functions but fails under load, lacks security, or requires extensive manual operation is unusable in the field.

Table 3.2: Non-Functional Requirements

NFR Category	Requirement Detail
Performance	System must respond to user queries within 2 seconds under normal load; critical alerts must propagate within 5 seconds.
Scalability	Architecture must support horizontal scaling to handle peak disaster-period traffic (10x normal load).
Reliability	99.5% uptime; automatic failover mechanisms; data replication across nodes.
Security	End-to-end encryption; JWT authentication; OWASP Top 10 compliance; ISO 27001 data handling.
Usability	Mobile-first responsive design; WCAG 2.1 accessibility; multilingual UI support.
Maintainability	Modular microservice-ready architecture; well-documented APIs; version-controlled codebase.
Interoperability	OpenAPI 3.0 compliant REST APIs; OAuth 2.0 for third-party integration.
Low-bandwidth Support	Core chatbot and reporting functions must operate at 2G speeds with partial offline caching.

3.3 Hardware Requirements

Table 3.3: Hardware Requirements

Component	Quantity	Specification
Development Workstations	4 units	Core i7 / Ryzen 7, 16 GB RAM, 512 GB SSD, Ubuntu/Windows 11
GPU for ML Training	1 unit (shared)	NVIDIA RTX 3060 / Google Colab Pro for model training
Cloud Server (prototype)	1 instance	AWS EC2 t3.medium or equivalent – backend deployment
Broadband Internet	Continuous	For API access (IMD, NASA, HDX) and collaborative development
Smartphone (testing)	2 units	Android 11+ for mobile interface and chatbot testing

3.4 Software Requirements

Table 3.4: Software Requirements

Category	Technologies / Tools
Frontend	React.js, HTML5, CSS3, JavaScript (ES6+), Leaflet.js
Backend	Node.js v18+, Express.js 4.x, REST API
AI / ML	Python 3.10, TensorFlow 2.x, PyTorch, scikit-learn, OpenCV 4.x
NLP	HuggingFace Transformers, NLTK, spaCy, CrisisNLP datasets
Database	PostgreSQL 15 + PostGIS extension, Redis (session/cache)
APIs / Data Sources	IMD Weather API, NASA Earthdata, HDX Humanitarian Data Exchange, CrisisNLP
DevOps / Tools	Docker, GitHub, VS Code, Jupyter Notebook, Postman
Authentication	JWT tokens, OAuth 2.0

3.5 Cost Estimation

The following table presents a cost estimate for the prototype implementation of CrisisCore. The project leverages open-source tools and academic API access to minimize expenditure.

Table 3.5: Cost Estimation

Item	Description	Estimated Cost
Cloud Hosting (Prototype)	AWS EC2 t3.medium × 3 months	₹ 2,500
Google Colab (ML training)	3 months subscription	₹ 0
Domain & SSL Certificate	1 year (prototype domain)	₹ 800
API Access (IMD, NASA)	Open/free academic access	₹ 0
Open-source Software Stack	React, Node, PostgreSQL, Python, TensorFlow	₹ 0
Development Hardware	Existing institutional infrastructure	₹ 0
Miscellaneous (storage, testing)	Dataset storage and misc tools	₹ 500
	TOTAL ESTIMATED COST	₹ 3,800

3.6 Assumptions

- Internet connectivity is available during system development and primary deployment; low-bandwidth modes are provided as fallback.
- Training datasets (CrisisNLP, HDX, historical IMD data) are publicly accessible for academic use.

- The platform will initially be deployed as a web application accessible via modern browsers; native mobile apps are planned for Phase II.
- Government agencies and NGOs participating in pilot testing will provide cooperation for integration and user feedback.
- Crowd-sourced reports from citizens are assumed to be largely genuine; spam filtering mechanisms are included in the design.

3.7 Risks and Challenges

3.7.1 Technical Risks

- **Data Quality:** Real-time data from citizen reports may be incomplete, inaccurate, or inconsistent. Mitigation: input validation, NLP-based filtering, and confidence scoring.
- **Model Generalization:** ML models trained on historical data may not generalize well to novel disaster patterns. Mitigation: regular retraining with updated data and ensemble approaches.
- **Scalability Under Disaster Load:** Platform may face extreme traffic spikes during active disasters. Mitigation: cloud auto-scaling and CDN integration.

3.7.2 AI Security Challenges

- **Adversarial Inputs:** Malicious actors may submit false reports to manipulate the system's risk assessment. Mitigation: anomaly detection and source reputation scoring.
- **Data Privacy:** Citizen location data and personal information must be protected. Mitigation: DPDPA 2023 compliance, data anonymization, and minimal data collection principles.
- **Model Bias:** Training data may reflect historical inequalities in disaster resource allocation. Mitigation: fairness auditing using AI Fairness 360 and diverse dataset curation.

3.8 Standard Policies and Compliance

CrisisCore is designed in compliance with the following national and international standards and policies:

- **Disaster Policies:** Disaster Management Act (2005), National Disaster Management Plan (NDMP 2019), NCRMP, and the Sendai Framework for Disaster Risk Reduction 2015–2030.

- Data & Security Standards: IT Act (2000), Digital Personal Data Protection Act (DPDPA 2023), OWASP Top 10, and ISO/IEC 27001.
- Software Engineering Standards: IEEE 29148 (Requirements Engineering), ISO/IEC 25010 (Software Quality), OpenAPI 3.0, and OAuth 2.0.
- Ethical Standards: Humanitarian Data Exchange (HDX) principles, Sphere Standards for humanitarian response, AI Fairness guidelines, and NITI Aayog Responsible AI Framework.

Chapter 4

Design

This chapter presents the complete design of CrisisCore, structured into three major sections: High Level Design (system architecture, module specifications, interface design, and database design), Detailed Module Design (AI/ML module internals), and Methodology (development approach and data collection strategy). The design is grounded in the requirements from Chapter 3 and the research insights from Chapter 2, and is guided by the principles of separation of concerns, modularity, scalability, and operational resilience under disaster conditions.

4.1 High Level Design

4.1.1 System Architecture

CrisisCore adopts a layered, microservice-compatible architecture comprising five primary tiers.

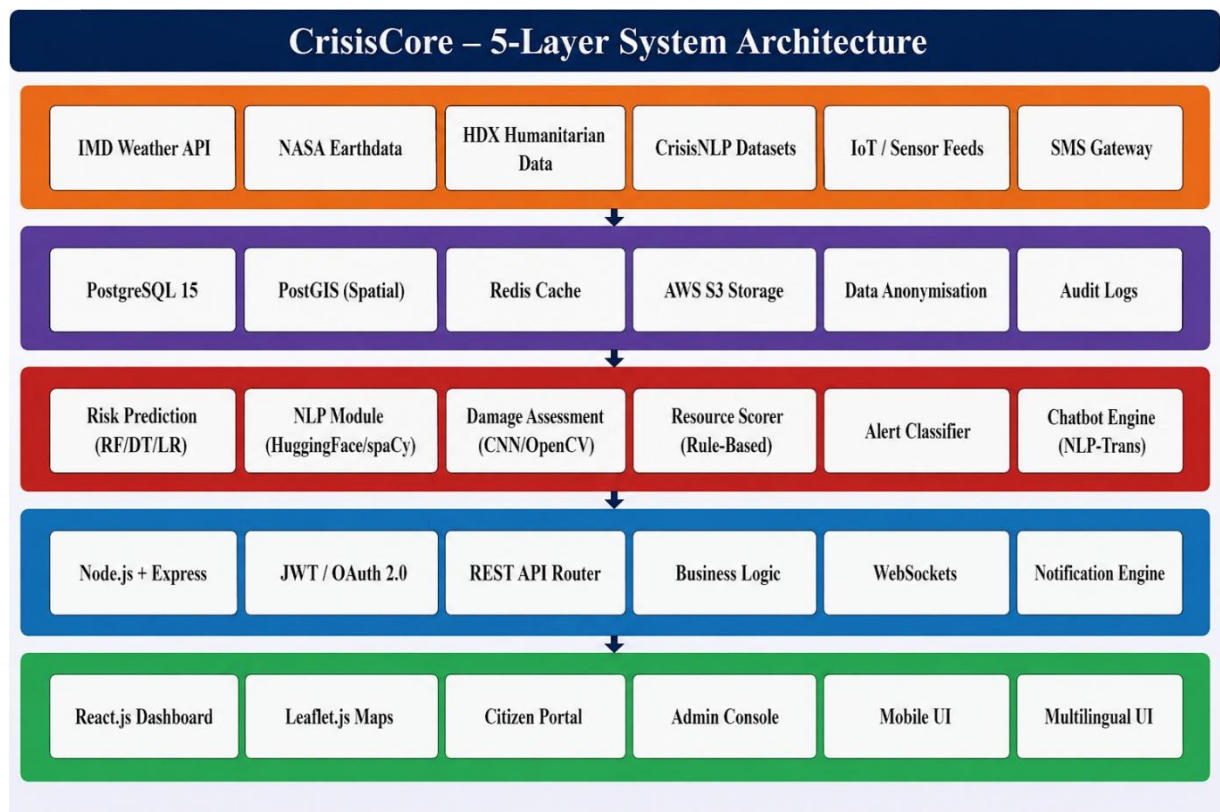


Figure 4.1: CrisisCore – 5-Layer System Architecture

- **Presentation Layer:** React.js-based responsive web application with Leaflet.js for geospatial map visualization. Supports role-based views for different user types (Admin, Government, NGO, Citizen).
- **API Gateway / Backend Layer:** Node.js + Express RESTful API server handling request routing, authentication (JWT/OAuth 2.0), and inter-module communication.
- **AI / ML Processing Layer:** Python-based services exposing REST endpoints for:
 - (a) Risk Prediction Engine (Decision Tree, Random Forest, Logistic Regression),
 - (b) NLP Module (Named Entity Recognition, text classification using CrisisNLP),
 - (c) Damage Assessment Module (OpenCV + TensorFlow CNN).
- **Data Layer:** PostgreSQL 15 with PostGIS extension for geospatial queries; Redis for session management and real-time alert caching; cloud storage (AWS S3) for media uploads.
- **Integration Layer:** Adapters for IMD Weather API, NASA Earthdata APIs, HDX Humanitarian Data Exchange, and CrisisNLP dataset streaming.

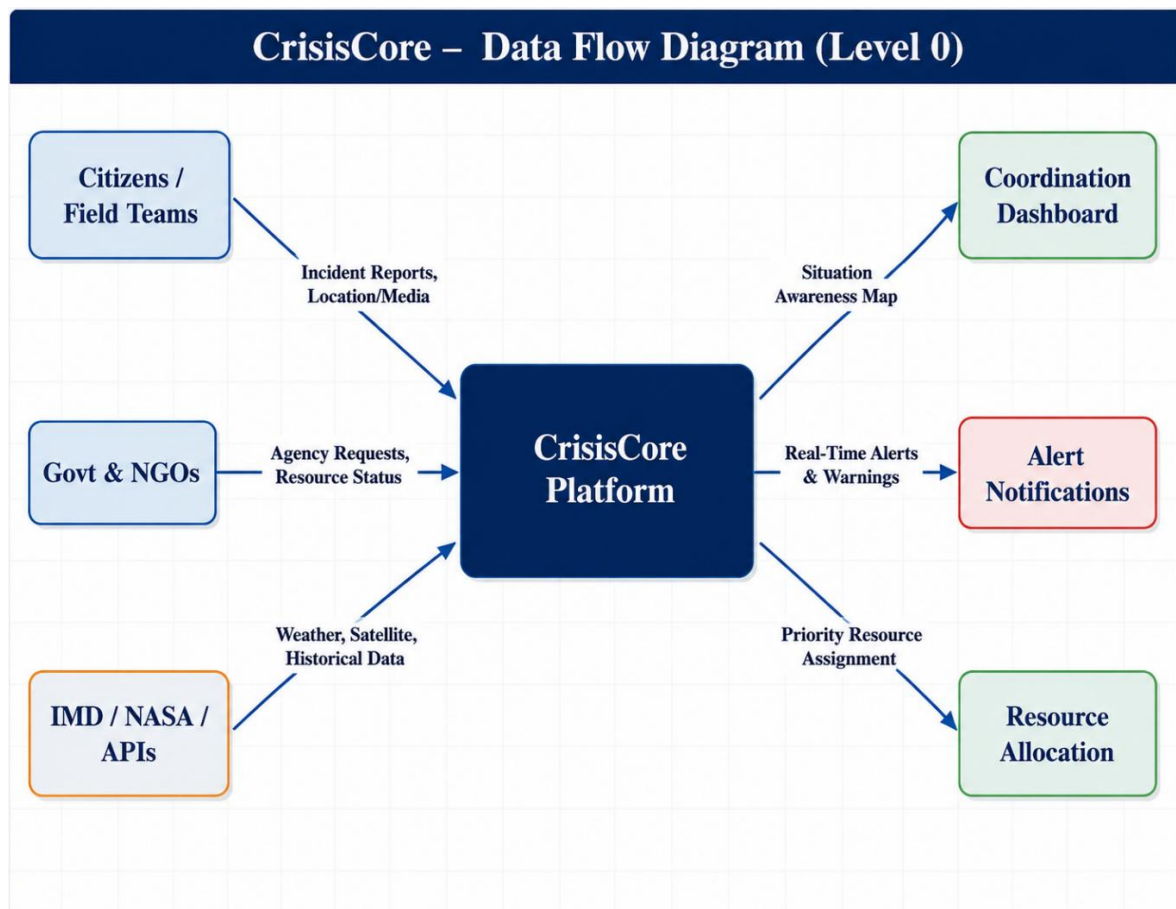


Figure 4.2: CrisisCore – Data Flow Diagram (Level 0)

4.1.2 Abstract Specification of Sub-Systems

The platform is decomposed into eight primary modules. Each module has clearly defined inputs, processing responsibilities, outputs, and REST API endpoints, enabling independent development, testing, and future extraction into microservices. Table 4.1 presents the abstract specification.

Table 4.1: Module Descriptions and Responsibilities

Module	Technology	Responsibility
Risk Prediction Engine	Python, scikit-learn, Random Forest	Predicts disaster risk zones using historical and real-time data; outputs risk scores and geographic heat maps; triggers early warning alerts.
NLP & Social Media Monitor	Python, HuggingFace, CrisisNLP	Classifies incoming text reports (social media + chatbot) using Naive Bayes / Logistic Regression; NER extracts location entities; keyword filtering for disaster events.
Damage Assessment Module	Python, OpenCV, TensorFlow CNN	Processes aerial/satellite imagery; detects damaged structures, flood extents, and road blockages; outputs severity scores.
Centralized Dashboard	React.js, Leaflet.js, Node.js	Real-time multi-role dashboard with live map, resource tracker, alert feed, incident log, and inter-agency communication panel.
Multilingual Chatbot	Node.js, NLP pipeline, translate API	Handles citizen emergency reporting in multiple Indian languages; guides users through emergency procedures; offline fallback mode.
Notification Engine	Node.js, SMS Gateway, Push Notifications	Dispatches real-time alerts to citizens and agencies via in-app notifications, SMS (Twilio / MSG91), and email.
Resource Allocation Engine	Node.js, Rule-Based Scoring	Maps available resources (hospitals, shelters, rescue teams) to disaster zones based on proximity, severity, and priority scores.
Database & GIS Layer	PostgreSQL + PostGIS, Redis	Stores all platform data with spatial indexing for location-based queries; Redis handles real-time session and alert caching.

4.1.3 Interface Design

The CrisisCore interface is designed following a mobile-first, accessibility-first philosophy. Key interface components include:

- **Coordinator Dashboard:** A full-screen map-centric view with overlaid disaster risk heat maps, live incident markers, resource availability panels, and alert management controls. Role-based data visibility ensures each agency sees only relevant information.

- **Citizen Reporting Interface:** A simplified, multilingual mobile-compatible form for submitting incident reports with photo upload, GPS-detected location, and severity selection. Designed for 2G network conditions.
- **Chatbot Interface:** A conversational UI accessible from the citizen portal, supporting text and structured quick-reply responses in multiple Indian languages.
- **Admin Console:** System health monitoring, user management, API status, and ML model performance dashboards.

Interface wireframes (Figure 4.3) follow WCAG 2.1 accessibility standards, ensuring screen-reader compatibility and high-contrast color schemes for use in field conditions. The design system is built on a component library derived from React + TailwindCSS with standardized spacing and typography scales.

4.2 Methodology

4.2.1 Research Design

CrisisCore follows an iterative, Agile development methodology structured around five phases: Requirements Elicitation (literature review + policy document analysis), Architecture Design (top-down decomposition + bottom-up technology validation), Prototype Development (sprint-based, core AI modules first), Model Development (ML pipeline per module with MLflow tracking), and Evaluation (unit → integration → end-to-end testing using simulated disaster scenarios).

4.2.2 Algorithms Used — Module-Wise

A. Disaster Risk Prediction — Weighted Ensemble (RF + DT + LR)

A three-classifier weighted ensemble is used. Random Forest ($B = 200$ trees) is the primary model, selected for its robustness to outliers and built-in feature importance via mean decrease in impurity. A single Decision Tree (max depth = 10) provides human-readable if-then rules for agency interpretability. Logistic Regression acts as a calibrated probability anchor with L2 regularisation ($C = 0.1$).

The final risk score is computed as:

$$\text{RiskScore}(\mathbf{x}) = 0.55 \times \mathbf{P_RF}(\mathbf{x}) + 0.25 \times \mathbf{P_DT}(\mathbf{x}) + 0.20 \times \mathbf{P_LR}(\mathbf{x})$$

Weights determined by 5-fold stratified cross-validation. Scores map to five classes: Very Low, Low, Medium, High, Critical. Alerts auto-trigger at High and Critical.

Eleven features are used as inputs: 3-day and 7-day cumulative rainfall, peak wind speed, soil saturation index, NDVI change rate, population density, hospital accessibility index,

road network density, historical disaster frequency (5-year), elevation gradient, and proximity to water body. Missing values are handled via k-NN imputation ($k = 5$); features are Min-Max normalised before RF/DT and standardised (zero mean, unit variance) before LR.

Evaluation metrics: Accuracy, Precision, Recall, macro F1-score, ROC-AUC, Expected Calibration Error (ECE), and SHAP feature attribution for interpretability.

B. NLP Pipeline — Text Classification + NER

Stage 1 — Text Classification: A **bert-base-uncased** transformer fine-tuned on the CrisisNLP dataset (80,000 samples, 8 classes: flood, earthquake, cyclone, landslide, fire, infrastructure failure, medical emergency, noise). Fine-tuning: AdamW optimiser, $lr = 2 \times 10^{-5}$, batch size = 32, 5 epochs with early stopping on validation F1. Achieves $F1 > 0.82$ on the CrisisNLP test set. BERT is chosen over Naive Bayes/TF-IDF because bidirectional attention captures the ambiguous, abbreviation-heavy language of crisis messages.

Non-English inputs are handled by **fastText** language identification (176-language lid.176.bin model, < 5 ms inference) followed by translation before classification.

Stage 2 — Named Entity Recognition: **spaCy en_core_web_lg** extended with custom disaster-domain entity types: DISASTER_LOCATION, INFRASTRUCTURE, CASUALTY_COUNT, and RESOURCE_REQUEST. Fine-tuned on 5,000 annotated disaster-domain sentences. Extracted locations are geocoded via OpenStreetMap Nominatim; entities with geocoding confidence < 0.65 enter a human-review queue rather than auto-creating incident records.

Evaluation: per-class and macro F1 for classification; entity-level Precision/Recall/F1 (strict match) for NER; median geocoding distance error (km) on a 500-sample evaluation set.

C. Damage Assessment — ResNet50 CNN (Fine-Tuned on xBD)

A **ResNet50** backbone pre-trained on ImageNet is fine-tuned on the **xBD benchmark** (22,068 pre/post-disaster satellite image pairs, 4-class pixel-level damage annotations: No Damage, Mild, Severe, Destroyed). ResNet50 is selected for its residual connections (stable fine-tuning of deep layers), strong accuracy/compute trade-off (25M parameters), and CPU-deployability at inference.

The ImageNet classification head is replaced with a task-specific two-layer head: Global Average Pooling $\rightarrow$ Linear(2048 $\rightarrow$ 512) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.5) $\rightarrow$ Linear(512 $\rightarrow$ 4) $\rightarrow$ Softmax. Class-weighted cross-entropy handles the imbalanced class distribution. Fine-

tuning uses a two-phase protocol: Phase 1 trains only the new head (5 epochs, $lr = 1 \times 10^{-3}$); Phase 2 unfreezes the last two ResNet blocks and fine-tunes end-to-end (15 epochs, $lr = 1 \times 10^{-4}$, cosine annealing).

Input images are tiled into 512×512 patches via OpenCV. Tile-level predictions are aggregated:

$$\text{damage_pct} = (\text{tiles classified Severe or Destroyed} / \text{total tiles}) \times 100$$

Evaluation: macro F1, Cohen's Kappa (vs. human annotators), and Intersection over Union (IoU) on a 500-image validation subset. Targets: macro F1 > 0.78, IoU > 0.72.

D. Resource Allocation — Weighted Priority Scoring

A deterministic weighted scoring algorithm is used (rather than ML) for full auditability — agency commanders can inspect and adjust weights without retraining. For each (incident, resource) pair:

$$\text{PriorityScore}(\mathbf{i}, \mathbf{r}) = 0.45 \cdot \text{Severity}(\mathbf{i}) + 0.35 \cdot \text{ProximityScore}(\mathbf{i}, \mathbf{r}) + 0.20 \cdot \text{CapacityScore}(\mathbf{r})$$

Where $\text{Severity}(\mathbf{i}) = \text{incident.severity} / 5.0$; $\text{ProximityScore} = 1 - (\text{ST_Distance} / D_{\text{max}})$, computed by PostGIS on WGS84 geometries with $D_{\text{max}} = 200$ km; $\text{CapacityScore} = \text{available_units} / \text{total_capacity}$. Resources are queried within radius using PostGIS ST_DWithin for spatial index acceleration. Status is updated to "reserved" immediately on assignment to prevent double-allocation.

E. Multilingual Chatbot — Intent Classification + FSM Dialogue

Intent classification uses **Multinomial Naive Bayes with TF-IDF features** (vocabulary: 5,000 unigrams + bigrams, Laplace smoothing $\alpha = 0.1$, trained on 3,000 annotated chatbot conversations across 12 intent classes). MNB is chosen over BERT here for inference speed (< 10 ms vs. 100–300 ms) — essential for 2G-compatible response times — since chatbot queries are short, structured, and less ambiguous than crisis tweets.

Dialogue is managed by an **8-state Finite State Machine**: GREETING → INTENT_IDENTIFIED → SLOT_FILLING → CONFIRM → ROUTE_TO_SERVICE / PROVIDE_INFO → ESCALATE → CLOSE. Transitions are deterministic, guaranteeing consistent, auditable behaviour critical for emergency interactions. If intent confidence < 0.40, the FSM transitions to ESCALATE (human handoff). Slot filling for required parameters (e.g., location for "find shelter") uses the spaCy NER pipeline from module B.

4.2.3 Data Collection

The following sources are used for training, validation, and real-time operation:

- **IMD Open Data APIs** — Real-time and historical weather data (rainfall, wind speed, temperature, soil moisture) at district level. Used for risk prediction features x_1 – x_3 .
- **NASA Earthdata (LANCE, SMAP, SRTM)** — Near-real-time satellite imagery (within 3 hours of overpass) for flood mapping and damage assessment; soil saturation (x_4); NDVI (x_5); elevation gradient (x_{10}).
- **HDX (Humanitarian Data Exchange)** — Population density (x_6), hospital accessibility (x_7), road density (x_8), and historical disaster impact datasets. Creative Commons licence for humanitarian use.
- **CrisisNLP Dataset (QCRI)** — 100,000+ labelled disaster tweets across 26 events; used for BERT fine-tuning and spaCy NER training.
- **National Disaster Database / NDMA Reports** — Indian district-level disaster event records 2000–2024; primary training labels for the risk prediction ensemble.
- **xBD Dataset (DIU)** — 22,068 pre/post-disaster image pairs with pixel-level damage annotations; used for ResNet50 fine-tuning.

All data collection adheres to DPDPA (2023) data minimisation and purpose limitation principles. Personal data is anonymised at ingestion. Training datasets are stored in AES-256 encrypted cloud storage with IAM access controls.

Chapter 5

Conclusion

5.1 Conclusion

CrisisCore represents a comprehensive, AI-driven response to the persistent challenges in disaster preparedness and response management. This Phase-I report has established the complete foundation for the platform: from a rigorous literature survey identifying critical research gaps, to a detailed requirements specification, and an end-to-end system architecture and design.

The proposed platform demonstrates how artificial intelligence, machine learning, NLP, and geospatial analytics can be integrated into a unified system that transforms disaster management from a reactive, manual process into a proactive, data-driven, and coordinated ecosystem. By enabling real-time situational awareness, predictive risk assessment, automated damage analysis, and inclusive multilingual citizen engagement, CrisisCore addresses the full spectrum of the disaster management lifecycle.

The design ensures scalability and extensibility: the modular architecture allows new AI models, data sources, and agency integrations to be added without disrupting the core system. Alignment with national policies (Disaster Management Act 2005, DPDPA 2023) and international frameworks (Sendai Framework, ISO 27001) ensures the platform's readiness for real-world regulatory environments.

Phase-I has successfully delivered: (a) an exhaustive literature survey with six key papers reviewed and research gaps clearly mapped to project objectives; (b) a complete requirement specification covering functional, non-functional, hardware, software, cost, assumptions, and risk dimensions; (c) a high-level system architecture, module specifications, and interface design; and (d) a clear methodology for model development and data collection.

Phase-II will focus on full system integration, model training and validation on real disaster datasets, end-to-end testing across simulated disaster scenarios, field evaluation with pilot agency partners, and preparation for real-world deployment across disaster-prone regions in India. The team is confident that CrisisCore has the potential to make a meaningful, measurable difference in disaster resilience and response efficiency.

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B.M.S. College of Engineering.
Department of AI & DS
Attainment of POs and PSOs

Batch No.: 8

Date: June 5th, 2026

Project Title: CrisisCore – A Real-Time AI Platform for Disaster Preparedness, Response & Coordination

PROGRAM OUTCOMES (POs)

PO	Level (3/2/1)	Justification (3=High, 2=Medium, 1=Low)
PO1	3	CrisisCore applies mathematics (probability models, statistical ML), computing fundamentals (algorithms, databases), and AI specialization to solve complex disaster prediction problems.
PO2	3	The project identifies, formulates, and analyzes the complex engineering problem of fragmented disaster response by reviewing research literature and reaching substantiated design conclusions.
PO3	3	The system designs a creative, multi-module solution for disaster coordination, considering public health, safety, and environmental impact in its architecture.
PO4	2	Disaster risk models are developed using research-based ML methods; data is analyzed and interpreted to validate model performance using standard metrics.
PO5	3	The project selects and applies modern engineering tools (TensorFlow, PostGIS, React, Node.js) and recognizes their limitations in real-world disaster scenarios.
PO6	2	Societal and environmental aspects (SDG alignment, rural accessibility, ethical AI) are analyzed in the design of the platform.
PO7	2	The project adheres to ethical AI principles (NITI Aayog AI framework, fairness auditing), data privacy laws (DPDPA 2023), and professional standards.
PO8	3	The project is carried out as a team of four, with defined roles and collaborative design reviews, simulating real-world engineering team dynamics.
PO9	3	The team has communicated findings effectively through review presentations and this formal technical report.
PO10	2	The project applies engineering management principles: structured work plan, sprint-based development, cost estimation, and risk management.
PO11	2	The project involves learning emerging AI/ML technologies, NLP, and geospatial frameworks, reflecting the team's adaptability and lifelong learning commitment.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO	Level (3/2/1)	Justification
PSO1	3	CrisisCore designs and develops a computing system based on core AI and Data Science principles: ML models, NLP pipelines, and spatial data processing.
PSO2	3	The project applies ML, data science, and AI tools (scikit-learn, TensorFlow, NLP) to address the real-world challenge of disaster response optimization.
PSO3	3	The team is building an end-to-end AI/DS system from requirements through architecture to prototype implementation, demonstrating full-stack interdisciplinary problem-solving.

Note: Level 3 = High (strongly addressed), Level 2 = Medium (moderately addressed), Level 1 = Low (minimally addressed).